



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 31.07.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : **CHEM F320**

Course Title : **Introductory Computational Chemistry Laboratory**

Instructor-in-Charge : **Prashant Uday Manohar** (pumanohar@pilani.bits-pilani.ac.in; #3268-P)

Co-Instructor : **Pinki** (p20230021@pilani.bits-pilani.ac.in; #3126/27)

1. Course Description: In this course the major focus is on practical computation of electronic structure of atoms and molecules using open source and proprietary software; specific computational experiments will be in the areas of potential energy surfaces, geometry optimization, molecular geometry from symmetry and trigonometric relations without visualization software, molecular orbitals and bonding patterns, Hartree-Fock calculations, correlation energy and size-consistency, DFT based calculations, computing excited states using CIS, EOM-CCSD jobs for computing energies of excited, ionized and electron-attached states; methods to estimate activation energy, solvent effects etc.; molecular dynamics simulation, molecular mechanics will also be explored; the actual experiments may vary and can have more specific learning outcomes so as to enhance the course with the latest developments in electronic structure theories of chemistry.

2. Scope and Objective of the Course: This is an elective laboratory course to gain introductory exposure to practical computational chemistry with major focus on computation of electronic structure of atoms and molecules. Other branches of computational chemistry such as molecular dynamics simulation, molecular mechanics will also be explored briefly.

3. Course Learning Outcomes:

- CLO1: Defining the computational problem and identifying the computational method to study the same
- CLO2: Preparing the input files and setting up the computational jobs
- CLO3: Analyzing the output file and the generated data and interpreting to understand the phenomena

4. Reference Books: Ira N. Levine, "Quantum Chemistry", (Pearson), A. Szabo and N. Ostlund, "Modern Quantum Chemistry" (McGraw-Hill); Donald MacQuarrie "Statistical Mechanics" (Viva Books); J. M. Haile, "Molecular Dynamic Simulation: Elementary Methods" (Wiley)

5. Course Plan:

Module No.	Title	Learning Objectives	Learning Outcome
1.	Introduction (2 experiments in 3 days)	Creating a molecular geometry without help of a visualization/front end software	Learn to generate Z-matrix and Cartesian coordinates using simple concepts in trigonometry
		Setting up a computational job for single point energy computation of polyatomic molecules using a front-end software and analyzing the output – Hartree-Fock energy, molecular orbital pictures, symmetry, etc. Koopmans' theorem would also be verified.	Setting up and analyzing an electronic structure computational job will be learned. The relationship of the analysis with the concepts in molecular symmetry, bonding, etc to be established.



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

Module No.	Title	Learning Objectives	Learning Outcome
2.	Potential energy and critical points (3 experiments in 3 days)	To obtain ground state potential energy curves and surfaces on two geometric parameters of a molecule – Study the bonding patterns and changes in various components of energy	Effect of geometry on the bonding and anti-bonding interactions will be studied using potential energy curves/surfaces
		To obtain ground state equilibrium structure of a molecule – to study to submit geometry optimization and harmonic frequency computation jobs, analyze the geometries and the frequencies, visualize them on IQmol.	The difference between a local minimum and saddle points to be understood from these two types of experiments
		To optimize a transition state structure – to study to submit transition state optimization job and analyze the output	
3.	Activation energy (2 experiments in 4 days)	To guess transition state of a simple reaction based on the geometries of reactant(s) and product(s), optimize the transition state and compute the activation energy of the reaction	To learn the practical aspects of freezing-string and growing string methods.
4.	Correlation energy and size-consistency (2 experiments in 3 days)	Setting up MP2, MP4 and CCSD jobs for a molecule and its non-interacting dimer	To learn how the total energy improves with use of sophisticated method
5.	Excited, ionized and electron attached states with and without electron correlation (4 experiments in 5 days)	To compute excited states using CIS and analyze the output	To learn to predict the character a few of low-lying excited states using Koopmans' theorem
		Setting up EOM-CCSD jobs for computing energies of excited, ionized and electron-attached states	Allowed versus forbidden transitions, ionization energy, electron affinity, etc
6.	Density functional theory based computation (2 experiments in 2 days)	Geometry optimization and computation of harmonic frequency for variety of exchange-correlation functionals.	To learn how Kohn-Sham DFT method works and get introduced to a few DFT exchange-correlation functionals.



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Laboratory practicals	2h-4h	75%	continuous	Open book
Mid-semester examination	1 day x 2 hr	10%	To be announced*	Closed book
Comprehensive examination	2 days x 2 hr	15%	To be announced*	Closed book

* The format of the mid-semester and comprehensive examinations will be announced well in advance.

Minimum Performance Requirement: Student missing *one or more complete modules* of practical sessions and/or the *mid-semester examination* and/or *comprehensive examination* may be reported as NC. Moreover, for obtaining a valid letter grade, the student *must score no less than 40% of the median* of the total performance.

7. Attendance and Make-up Policy: Attendance is compulsory. Students must perform all the experiments and submit the records regularly. Due to operational reasons, there is only limited scope for offering make-up for any experiment. Make-ups may be considered only in exceptional circumstances.

8. Chamber Consultation Hour: To be announced.

9. Notices: Notices and announcements related to the course will be shared with the registered students through nalanda or erp portal or email.

Instructor-in-charge
Course No. CHEM F320